Latent Space Representation of Coding Experience

This project implements a single-layer perceptron to classify students based on their coding experience across six programming languages: C/C++, Python, MATLAB, R, Assembly, and Java.

Project Structure

perceptron.ipynb: The main notebook containing the data generation, perceptron implementation, and visualization code.
*data/: The project uses synthetic data generated directly by the notebook code.

How to Run

1. Open the project in Google Colab.
2. Run all cells sequentially. The code will generate a dataset of 1000 student entries.
3. The model will perform classification and generate a 2D plot of the decision boundary for Python vs. C/C++ experience.

Project Objectives

The project follows these steps to implement a linear classifier:
Data Preparation: Creating a 1000-entry dataset representing lines of code per language.

Experience Function: Designing a scoring function to aggregate coding proficiency.

Perceptron Model: Implementing a neural network architecture ($w^Tx \ge b$) to differentiate students who have taken a programming class from those who have not.

*Visualization: Drawing the vector w and the decision boundary $w^Tx = b$ to visualize the linear partition in 2D space.

Methodology

The perceptron algorithm iterates through the data and updates the weight vector w whenever a classification error occurs ($w \leftarrow w + y_i x_i$). This continues until the data is partitioned correctly, or the maximum number of iterations is reached.